ENTERPRISE DATA GOVERNANCE FRAMEWORK DESIGN & IMPLEMENTATION FOR E-COMMERCE PLATFORM TRANSFORMATION

DATA GOVERNANCE TARGET OPERATING MODEL (TOM)

Document status: Proposed design for an illustrative e-commerce transformation case study. Organizational roles, thresholds, SLAs, retention periods, and regulatory interpretations require validation before production adoption.

Document owner: Enterprise Data Governance | Version: 2.0 | Review cycle: Annual and upon material business, technology, or regulatory change

1. EXECUTIVE SUMMARY

This Target Operating Model (TOM) describes how enterprise data governance will be directed, delivered, measured, and continuously improved across an e-commerce platform. It adopts a federated model: enterprise governance sets common policy, standards, decision rights, and controls, while business domains remain accountable for the meaning, quality, appropriate use, and lifecycle of their data. Technology teams implement and operate the technical controls.

The model covers Customer, Product, Order & Transaction, and Vendor/Supplier domains, with cross-functional participation from Marketing, Merchandising, Operations, Finance, Payments, Supply Chain, Data/Analytics, IT, Security, Privacy, Legal, and Risk/Compliance.

2. PURPOSE, OUTCOMES, AND DESIGN PRINCIPLES

- Establish accountable ownership and clear decision rights for critical data assets.
- Improve fitness-for-purpose data quality and reduce avoidable business, customer, financial, fraud, and reporting impacts.
- Make data discoverable and understandable through a governed catalog, business glossary, technical metadata, and lineage.
- Embed privacy, security, retention, and compliance controls throughout the data lifecycle.
- Enable trusted reporting, personalization, experimentation, fraud/risk analytics, and AI/ML use cases.
- Scale consistently while allowing domain-level execution within enterprise standards.

3. SCOPE AND ASSUMPTIONS

- Data domains: Customer; Product/Catalog; Order, Payment & Transaction; Vendor/Supplier.
- Systems and flows: web/mobile storefront, CRM, product information/catalog tools, order management, payment service providers, logistics/fulfilment, ERP/finance, APIs/events, operational databases, data lake/lakehouse, data warehouse, BI and ML platforms.
- Governance applies to structured and unstructured data, derived attributes, reports, data products, analytical datasets, and relevant third-party data.
- The CDO role is assumed for the target state. If no CDO exists, an executive sponsor or equivalent accountable leader must hold the enterprise mandate.
- Tools listed in this TOM are examples, not mandated vendor selections.

4. FEDERATED OPERATING MODEL

✓ **Governance Model Type: Federated Governance Model**

Layer	Core Participants	Accountability / Decision Focus	Typical Cadence
Executive direction	Executive sponsor/CDO; Data Governance Council	Approve policy, risk appetite, priority domains, material exceptions, funding, cross-domain decisions and escalations.	Quarterly; ad hoc for critical decisions
Enterprise enablement	Data Governance Lead/Office; Enterprise/Data Architects; Privacy, Security, Legal, Risk/Compliance	Maintain standards, glossary conventions, governance controls, reporting, change process, training and assurance.	Monthly working group
Domain accountability	Data Owners; domain business leads	Approve domain definitions, critical data elements, quality targets, permitted uses and remediation priorities.	Monthly domain review
Operational execution	Data Stewards; Data Product Owners; analysts; process owners	Maintain definitions and metadata, monitor quality, investigate issues, coordinate fixes and validate outcomes.	Weekly or operationally appropriate

Technical enablement	Data Custodians, Data Engineering, Platform/DBA, application owners, Security Operations	Implement access, validation, logging, lineage, encryption, backup, retention, deletion and platform controls.	Continuous operations
Independent challenge	Internal Audit / assurance functions	Assess design and operating effectiveness independently; do not own day-to-day control execution.	Risk-based plan

5. GOVERNANCE STRUCTURE AND ROLE CHARTERS

5.1. Executive Sponsor / Chief Data Officer (CDO)

- Sponsors the enterprise data strategy and secures executive support, capacity, and funding.
- Chairs or delegates chairing of the Data Governance Council; resolves issues beyond domain authority.
- Ensures governance priorities align with business strategy, risk appetite, and regulatory obligations.
- Accountable for the enterprise governance operating model, not for personally performing all operational controls.

5.2. Data Governance Council

- Membership: CDO/executive sponsor, domain business leaders, IT/Data/Architecture, Security, Privacy, Legal, Risk/Compliance, and Finance/Operations as needed.
- Approves enterprise policies and standards; endorses domain-level standards where cross-domain impact exists.
- Prioritizes critical data elements and remediation investments; resolves conflicting definitions and ownership disputes.
- Reviews KPI trends, material data risks, exceptions, audit findings, and overdue high-severity issues.
- Records decisions, rationale, accountable action owners, due dates, and dissent/escalation where applicable.

5.3. Data Governance Lead / Office

- Maintains the governance framework, policy inventory, standards, templates, glossary conventions, control register, and KPI definitions.
- Coordinates Council meetings, decision logs, maturity assessments, domain onboarding, communications, and adoption.
- Monitors implementation progress and challenges incomplete or inconsistent governance evidence.
- Coordinates but does not replace Data Owners' accountability or Security/Privacy specialists' formal responsibilities.

5.4. Data Owner

- A named business leader accountable for a defined data domain or data product.
- Approves business definitions, critical data elements, quality rules/targets, access purpose, acceptable use, and remediation priority.
- Accepts residual domain data risks only within delegated authority and approved risk appetite.
- Ensures stewards and process owners have time and authority to execute governance tasks.

5.5. Data Steward

- Maintains business glossary terms, data dictionaries, classifications, metadata, and quality-rule documentation.
- Monitors scorecards, triages data issues, coordinates root-cause analysis, tracks remediation, and validates business outcomes.
- Coordinates with upstream source owners and downstream data consumers; escalates missed SLAs and disputed definitions.
- Does not unilaterally approve material policy exceptions unless explicitly delegated.

5.6. Data Custodian / Technical Owner

- Implements and operates technical controls for storage, processing, access, encryption, backup, logging, retention, deletion, and recovery.
- Maintains technical metadata, pipeline controls, schema/change records, and technical lineage where feasible.
- Supports incident response, impact analysis, quality-rule automation, and evidence collection.
- Works with Security and Privacy specialists to ensure technical controls meet approved requirements.

5.7. Supporting and Control Functions

- Privacy Officer/Privacy function: advises on lawful processing, notices, data subject rights, minimization, privacy impact assessment, and personal-data incident obligations.
- Information Security: defines security control requirements, threat response, privileged access safeguards, encryption standards, and security incident handling.
- Legal and Risk/Compliance: interpret applicable legal, contractual, payment, consumer, tax, and sector-specific obligations.
- Enterprise/Data Architect: defines data architecture principles, integration patterns, canonical models, and design/change review.
- Data Product Owner / Analytics Lead: defines data-product service expectations, consumer needs, and fitness-for-purpose acceptance criteria.
- Internal Audit: independently tests governance and controls; remains separate from operational ownership.

6. DOMAIN ACCOUNTABILITY AND CRITICAL DATA EXAMPLES

Domain	Proposed Business Owner	Stewardship Focus	Illustrative Critical Data Elements
Customer	CRM/Marketing or Customer Operations leader	Identity resolution, consent/preference metadata, customer status, contactability, duplication and permitted use.	customer_id, email/phone, consent status, region/country, customer status
Product/Catalog	Merchandising/Catalog leader	SKU uniqueness, product hierarchy, attributes, pricing validity, availability and publication status.	product_id/SKU, category, price, currency, product status, seller/vendor association
Order & Transaction	Operations/Order Management leader with Finance/Payments participation	Order state transitions, payment reconciliation, refunds, timestamps, amounts and auditability.	order_id, payment_status, transaction_id, amount, currency, refund_status, timestamps
Vendor/Supplier	Supply Chain/Procurement leader	Supplier identity, onboarding status, payment details, contractual status and duplicate vendor prevention.	vendor_id, legal name, tax/registration identifier where applicable, status, payment master data

Owner assignments above are illustrative. The organization must name individuals or formally delegated positions and identify a source system of record for each critical element.

7. RACI FRAMEWORK

R = Responsible for execution; A = Accountable for the outcome/decision; C = Consulted; I = Informed. Each activity should have one clearly identified A. Security, Privacy, Legal, Architecture, and Audit may have additional control or review obligations not fully represented in this simplified matrix.

Activity	Council	Governance Lead	Data Owner	Steward	Custodian / IT	Control Functions
Approve enterprise policy	A	R	C	I	C	C
Approve domain definitions/CDEs	I (escalation)	C	A	R	C	C where relevant
Define and approve DQ rules/targets	I	C	A	R	C	C where relevant
Monitor DQ and manage issues	I (material issues)	C	A	R	R for technical fix	C
Approve business access purpose	I	I	A	C	R for provisioning	C/approval per security policy
Implement technical access/security controls	I	C	C	I	R	A/R for security standards as applicable
Maintain catalog, glossary and lineage	I	A for framework	A for domain content	R	R for technical metadata	C
Approve high-risk policy exception	A or delegated executive	R for workflow	C/A within delegated authority	I	C	C/A for control sign-off
Resolve cross-domain definition conflict	A	R to coordinate	C	C	C	C
Independent control assurance	I	C	C	I	C	Internal Audit: A/R for audit work

RACI refinement: the original matrix treated the Governance Council as informed on many operational activities and used 'Policy Approval: Owner=R, Council=A'. The revised approach retains Council accountability for enterprise policy approval, while domain Owners are accountable for domain-level decisions. Access approval should distinguish business-purpose authorization from technical provisioning and security approval.

8. CORE GOVERNANCE PROCESSES

8.1. Critical Data Element (CDE) Management

1. Identify data elements that materially affect revenue, payments, customer outcomes, fraud/risk, regulatory reporting, or critical operations.
2. Record business definition, owner, steward, source of record, classification, consumers, lineage, quality rules, and consequences of failure.
3. Approve CDEs and target thresholds through the relevant Data Owner; escalate cross-domain impacts to the Council.
4. Review CDE inventory after major product, platform, process, vendor, or regulatory changes.

8.2. Data Quality Management

- Use fit-for-purpose dimensions: accuracy, completeness, validity, consistency, uniqueness, timeliness, and integrity where relevant.
- Define rule logic, denominator, exclusions, threshold, severity, monitoring frequency, rule owner, source, and remediation action for each rule.
- Monitor at source and in analytical pipelines where appropriate; prevent known-invalid critical data from silently propagating.
- Track issues from detection through triage, root cause, remediation, independent/business validation, closure, and recurrence prevention.
- Report both score and impact; a high aggregate score must not hide a critical payment, privacy, or customer-impacting defect.

8.3. Metadata, Catalog and Lineage

- Register critical datasets and data products with business definition, owner, steward, classification, source, refresh frequency, quality status, consumers, and support contact.
- Maintain business glossary terms and agreed definitions for shared metrics such as gross merchandise value, net sales, cancelled order, active customer, and refund.
- Document end-to-end lineage for Tier-1 datasets from source through ingestion/transformation to reporting or model consumption; identify manual transformations and material dependencies.
- Perform impact analysis before material schema, transformation, definition, retention, or access changes.

8.4. Access, Privacy and Security

- Use least privilege and role-based access, with attribute/context-based controls where required by sensitivity, geography, purpose, or business context.
- Separate business approval of purpose from technical provisioning; enforce segregation of duties for privileged access.
- Use approved authentication, encryption, masking/tokenization, logging, monitoring, and periodic access recertification.
- Assess new or materially changed uses of personal data, including personalization, profiling, AI/ML, and third-party sharing, with Privacy/Legal/Security as appropriate.
- Maintain evidence of access decisions, data sharing approvals, and periodic reviews.

8.5. Master Data Management

- Define authoritative sources, matching/merging rules, survivorship logic, golden-record ownership, exception handling, and audit history for Customer, Product, and Vendor master data.
- Do not assume a single physical database is always required; the 'single source of truth' means an agreed authoritative record and governed distribution pattern.
- Protect source history and ensure merges, unmerges, and identity resolution can be traced and corrected.

8.6. Data Classification

Define classification levels:

- Public
- Internal
- Confidential
- Restricted (PII / highly sensitive)

9. DATA ISSUE SEVERITY, SLA AND ESCALATION

Severity	Illustrative Criteria	Initial Response Target	Resolution Target	Escalation
Critical	Active privacy/security exposure; materially incorrect payment/financial data; widespread customer or order impact; regulatory reporting risk.	Within 4 hours	Within 24 hours or approved containment plan	Immediate to Owner, Security/Privacy/Legal as applicable, Governance Lead and executive sponsor
High	Significant domain impact, broad reporting defect, material reconciliation issue, or key data product unavailable.	Within 1 business day	Within 5 business days	Owner and Governance Lead
Medium	Limited impact with workaround; localized quality or metadata issue.	Within 2 business days	Within 10 business days	Domain Owner if overdue
Low	Minor defect, documentation gap, or low-impact improvement.	Within 5 business days	Within 30 business days	Domain backlog review

10. TECHNOLOGY AND CONTROL ARCHITECTURE

- **Source systems**: storefront, CRM, catalog/PIM, order management, payment providers, logistics, ERP/finance, supplier systems and external feeds.
- **Integration**: API, event streaming, batch ingestion, CDC, and controlled file exchange with validation and reconciliation.
- **Data platform**: operational stores, lake/lakehouse, warehouse, semantic/metrics layer, BI and ML environments.
- **Cross-cutting governance services**: catalog/glossary, lineage, classification, access policy, quality rules/scorecards, MDM, audit logs, retention/deletion controls, and issue tracking.
- Controls should be automated where proportionate and feasible; manual controls must have named owners, frequency, evidence, and exception handling.

- **Tool** choices such as Microsoft Purview, Collibra, Alation, Atlan, Great Expectations, Informatica, Snowflake, BigQuery, or Databricks are examples subject to architecture, security, cost, integration, and procurement assessment.

11. KPI FRAMEWORK AND DEFINITIONS

KPI	Formula (illustrative)	Initial Target	Steady-State Target	Owner / Cadence
CDE Data Quality Pass Rate	CDE records passing all applicable approved rules / CDE records tested × 100	≥90%	≥95%	Data Owner/Steward; weekly or monthly
Critical Asset Ownership Coverage	Critical assets with named Owner / total critical assets × 100	≥95%	100%	Governance Lead; monthly
Metadata Completeness	Critical assets meeting mandatory metadata fields / total critical assets × 100	≥80%	≥98%	Governance Lead/Stewards; monthly
Lineage Coverage	Tier-1 datasets with approved end-to-end lineage / total Tier-1 datasets × 100	Prioritized pilot	100% target	Architecture/Data Engineering; monthly
Issue SLA Compliance	Issues resolved within applicable SLA / issues due for resolution in period × 100	≥80%	≥95%	Domain Owner; monthly
Access Review Completion	Required reviews completed on time / reviews due × 100	≥90%	100% for privileged access	Security/System Owners; quarterly
Recurring Issue Rate	Issues recurring from a previously closed root cause / total closed issues × 100	Establish baseline	Downward trend	Stewards; monthly

Critical Data Incident Trend	Count and severity-weighted trend of material data incidents	Establish baseline	Sustained reduction	Governance Council; quarterly

Targets should be baselined and approved. Define whether measurements are record-weighted, rule-weighted, dataset-weighted, or business-impact-weighted. Avoid claiming that a target has been achieved until supported by measured evidence.

12. IMPLEMENTATION ROADMAP

Phase	Key Activities	Exit Criteria / Evidence
1. Assess & mobilize	Confirm sponsor, scope, stakeholders, current-state maturity, system inventory, risks, and priority use cases.	Approved charter, domain inventory, gap assessment, initial risk register.
2. Design foundation	Approve TOM, policy, RACI, classification, CDE criteria, glossary standards, issue workflow, KPI definitions.	Signed-off TOM/policy, named owners, decision log, prioritized control backlog.
3. Pilot high-value domains	Pilot Customer and Product or Order/Payment based on risk; implement catalog, DQ rules, access reviews, lineage and issue tracking.	Baseline and scorecards, evidence of rule execution, resolved issues, lessons learned.
4. Scale and integrate	Extend to remaining domains, systems, data products and third parties; automate controls and lineage.	Domain onboarding completed, agreed coverage targets, operating procedures and training.
5. Optimize and assure	Review trends, test control effectiveness, refresh risk assessment, simplify recurring controls and improve adoption.	Quarterly governance review, action plan, independent assurance results, updated maturity assessment.

13. FORUMS, ARTEFACTS AND CADENCE

Forum / Artefact	Minimum Content	Cadence / Accountable Party
Data Governance Council	Decisions, material risks, cross-domain conflicts, policy exceptions, investment priorities, KPI trends.	Quarterly; Council Chair
Domain Data Review	DQ scorecard, CDE changes, open issues, access/use concerns, metadata and lineage gaps.	Monthly; Data Owner
Operational issue review	Severity, root cause, owner, SLA, remediation, validation, recurrence prevention.	Weekly or as needed; Steward
Policy and standards register	Version, owner, approver, effective date, review date, exceptions and related controls.	Continuous; Governance Lead
Data catalogue / glossary	Definitions, owners, classification, source, lineage, consumers, quality status.	Continuous; Steward/technical teams
Control evidence register	Control objective, frequency, executor, evidence, exceptions, remediation and reviewer.	Per control cycle; control owner

14. INTEGRATION WITH DATA ARCHITECTURE

- Governance embedded in:
  - Data ingestion pipelines
  - Transformation layers
  - BI / reporting layer
- Controls applied at:
  - Data creation
  - Data storage
  - Data access

15. COMPLIANCE & RISK MANAGEMENT

- Align with regulatory frameworks (e.g., GDPR principles)
- Implement audit trails
- Periodic compliance assessment
- Risk identification and mitigation

16. ESCALATION FRAMEWORK

- Data issues escalated:
    - Steward → Owner → Governance Council
- SLA for resolution defined

  SLA Table (Example):

Severity	Initial Response	Target Resolution
Critical	< 4 jam	< 24 jam
High	< 1 hari kerja	< 5 hari kerja
Medium	< 2 hari kerja	< 10 hari kerja
Low	< 5 hari kerja	< 30 hari kerja

17. ADOPTION, TRAINING AND CONTINUOUS IMPROVEMENT

- Provide role-based onboarding for Owners, Stewards, Custodians, analysts, product teams, and executives.
- Assign governance champions in each domain and embed responsibilities into existing business and technology processes.
- Use a quarterly improvement cycle: measure, prioritize root causes, implement fixes, validate effectiveness, and update standards.
- Reassess TOM annually and after major organizational, platform, acquisition, product, or regulatory changes.

18. APPENDICES

Appendix A: Sample Data Issue Workflow

1. Detect through validation, reconciliation, user report, monitoring, audit, or incident alert.
2. Log issue with dataset/element, evidence, impact, severity, date detected, source, and suspected cause.
3. Triage and assign the Steward; notify Owner and control functions if risk warrants.
4. Contain immediate harm where necessary, including pausing a feed, restricting access, or correcting a downstream report.
5. Perform root-cause analysis and assign a corrective action to the process/system owner.

6. Implement fix, test source and downstream impacts, and obtain business validation.
7. Close only when evidence is recorded; add prevention controls and monitor recurrence.

Appendix B: Decisions Required Before Production Adoption

- Confirm executive sponsor/CDO equivalent and Council membership.
- Name accountable Data Owners and Stewards for every in-scope domain and critical asset.
- Approve actual thresholds, SLA clocks, severity definitions, data retention schedule, and control evidence requirements.
- Validate legal, privacy, consumer, payment, tax, contractual, and cross-border data obligations with qualified internal counsel/compliance specialists.
- Select tooling only after capability, architecture, security, privacy, cost, and operating-model assessment.